

Ryan Cooley

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GitHub: <https://github.com/Ryan-Cooley> | Portfolio: <https://ryan-cooley.github.io/RCPortfolio/>

EDUCATION

Tufts University

Medford, MA

Bachelor of Science in Chemical Physics; Minor in Applied Computational Science

Expected May 2027

GPA: 3.89/4.0 | Dean's List (All Eligible Semesters) | Sigma Pi Sigma (Physics Honor Society)

Relevant Coursework: Physical Chemistry, Computational Biology, Quantum Theory, Probability, Linear Algebra

SKILLS

Programming Languages: Python, C++, SQL, JavaScript

ML & Data: pandas, NumPy, Matplotlib, Numba (JIT), PyTorch, RDKit, Gemmi, Biopython

Computational Chemistry: Potts/statistical energy models, MSA curation (Jackhmmer), ColabFold, OpenMM, VMD

Backend & Web: PostgreSQL, FastAPI, React

Infra & DevOps: AWS (SageMaker, S3, EC2), Linux, Bash, Docker, systemd, Slurm, Git, GitHub Actions, pytest

EXPERIENCE

Takeda Pharmaceutical Company Limited

Billerica, MA

Machine Learning Pipeline Development Intern

May 2026 – Aug 2026

- Shipped a production benchmarking platform for protein structure prediction models on AWS (SageMaker, EC2, S3), spanning a normalized PostgreSQL results store, an S3 backed data layer, a scoring pipeline, a FastAPI service, and a React frontend, deployed live and serving ranked results to the team
- Designed a concurrent scoring queue in PostgreSQL (FOR UPDATE SKIP LOCKED) running as systemd worker services, parallelizing evaluation across 8 vCPUs on two concurrency axes with zero double processing
- Built a metrics pipeline that generalizes across prediction tasks using versioned schema migrations, a metric metadata registry, and a metadata-driven API and UI, so onboarding a new task requires no changes to serving code
- Developed FastAPI endpoints (leaderboard, per-sample results, CSV export, presigned-URL uploads) and a 5-step React upload wizard that ingests an S3 folder and enqueues models for scoring

Ding Lab at Tufts University

Medford, MA

Undergraduate Research Associate

May 2024 – Present

- Applied a lab-developed EM-based 2-component Potts mixture model to a Jackhmmer MSA of GA/GB streptococcal protein G homologs, partitioning sequences into fold-directed sub-alignments via per-sequence log-probability ratios with no structural labels
- Validated the partition structurally and by contact prediction: a Potts-identified sequence folded to ~ 2 Å C α RMSD vs. the GA crystal (~ 13 Å vs. GB) and flipped to ~ 3.8 Å vs. GB on the GB-directed sub-MSA (ColabFold, 3 seeds); ESM-MSA-Transformer contacts scored $\sim 10\times$ the random-MSA baseline, confirming fold-discriminating couplings without structural supervision
- Built an automated multi-ligand evaluation framework with Boltz-2, Gemmi, RDKit, and Biopython on Tufts HPC (Slurm); assessed ligand placement accuracy via RMSD against experimental PDB structures across the HIF2-alpha and EG5 systems

Entegris

Billerica, MA

Metrology Retention Intern (Data Automation and Simulation)

May 2025 – Aug 2025

- Automated reporting with VBA macros, cutting workload by 90% and saving 3+ hours weekly
- Improved analysis speed from 38 minutes to under 3 minutes, enabling same-day client turnaround
- Built a prototype particle-tracking simulation to visualize membrane transport for clients

PROJECTS

Open Source - GitHub (Ryan-Cooley)

Remote

High-Performance Simulation Engine: Monte Carlo Option Pricer (Python / Numba)

Jun 2025 – Present

- Build a Numba Monte Carlo pricer: $10\times$ vs NumPy on 1M+ paths; $RMSE \leq 5e-4$ (vs Black-Scholes)
- Develop IV calibration (Brent/bisection, no-arbitrage) and a delta-hedging simulator with costs; map cost-error frontiers and log P&L
- Ship a reproducible harness with CLI: seeded RNG, profiling, unit tests, automated benchmarks (CSV/plots)

LEADERSHIP

Tufts SEDS - CubeSat Team

Medford, MA

Communications & Ground Station Lead

Sep 2024 – Present

- Assess uplink/downlink protocols for CubeSat operations, leveraging FCC Amateur Radio Technician License
- Perform orbital mechanics analysis; simulate trajectories with “42” and model space debris with MASTER